CAM-IMX577-12MP User Manual

Date	Version
2025/12/06	V1.0

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1. Product Description

1.1 Overview

The CAM-IMX577-12MP is a high-performance MIPI CSI-2 camera module featuring the Sony IMX577 12.3-megapixel back-illuminated CMOS image sensor, designed to deliver exceptional 4K imaging capabilities with high dynamic range and low-light performance.

This module is supported on the NVIDIA Jetson Orin Nano platform and Raspberry Pi 5. It includes a factory-optimized tuning file and is paired with a high-quality lens, providing a wide field of view (HFOV: 97.43°, VFOV: 80.97°, DFOV: 109.82°), making it suitable for demanding vision applications.

The module leverages the existing IMX477 driver in JetPack 6.2, eliminating the need for separate driver installation; we provide guidance on configuration via the Jetson-IO tool.

Under the latest NVIDIA JetPack SDK, it supports full resolution 4056 × 3040 (true 12MP) in RAW10 format, ideal for AI vision, high-resolution surveillance, industrial AOI, machine vision, scientific imaging, and long-term time-lapse projects on the Jetson Orin Nano.

1.2 Key Features

- Genuine Sony STARVIS IMX577 sensor — real hardware 12MP with 1/2.3” optical format
- Supported resolutions: 4056 × 3040 (full resolution), 3840 × 2160 (4K UHD), 1920 × 1080 (Full HD)
- Starlight-level full-color imaging down to 0.01 Lux
- Wide lens field of view: HFOV 97.43°, VFOV 80.97°, DFOV 109.82°
- Built-in IR filter
- Native support through NVIDIA Argus / GStreamer (leverages IMX477 driver in JetPack 6.2)
- Full 4-lane MIPI CSI-2 bandwidth utilization on Jetson Orin Nano
- DOL-HDR up to 110 dB
- Low power consumption, engineered for 7 × 24 h continuous operation

2. Technical Specifications

2.1 Sensor

Item	Specification
Sensor Model	Sony IMX577 (STARVIS series)
Sensor Type	1/2.3″ Back-illuminated stacked CMOS
Total Pixels	4056 (H) × 3040 (V) = 12.3 Megapixels
Pixel Size	1.55 μm × 1.55 μm
Shutter Type	Electronic rolling shutter
Output Format	RAW10 (default) / RAW12
Max SNR	37.1 dB
Dynamic Range	71.3 dB (standard), up to 110 dB (DOL-HDR)

2.2 Supported Resolutions

Raspberry Pi 5 (libcamera driver)

Pixel Format	Resolution	Description
SRGGB12_CSI2P	4056 × 3040	12-bit RAW Full Resolution
SRGGB12_CSI2P	4056 × 2160	12-bit RAW 4K UHD
SRGGB12_CSI2P	2028 × 1520	12-bit RAW 2K
SRGGB12_CSI2P	2028 × 1080	12-bit RAW Full HD
SRGGB12_CSI2P	1332 × 990	12-bit RAW Low Power

NVIDIA Jetson Orin Nano (JetPack 6.2, IMX477 driver)

Resolution	Description
4032 × 3040	Full Resolution RAW10
3840 × 2160	4K UHD RAW10
1920 × 1080	Full HD RAW10

2.3 Lens Optics

Item	Specification
Horizontal FOV (HFOV)	97.43°
Vertical FOV (VFOV)	80.97°
Diagonal FOV (DFOV)	109.82°

2.4 Electrical

Item	Specification
Interface	MIPI CSI-2 (4-lane on Jetson Orin Nano)
Power Supply	3.3 V (directly from Jetson CSI port)
Operating Current	≈220 mA

2.5 Mechanical

Item	Specification
Module Dimensions	38 × 38 mm
Weight	≈12–15 g (including flex cable)
Flex Cable	150 mm 15-pin 1.0 mm + 150 mm 22-pin 0.5 mm
Mounting Holes	4 × M2, compatible with Jetson cases & mounts

2.6 Environmental

Item	Specification
Operating Temperature	–20 °C to +70 °C
Storage Temperature	–30 °C to +85 °C
Humidity	20%–85% RH (non-condensing)

3. Hardware

3.1 Camera Module Pin Out

Pin #	Name	Description
1	GND	Ground
2	CAM_D0_N	MIPI Data Lane 0 Negative
3	CAM_D0_P	MIPI Data Lane 0 Positive
4	GND	Ground
5	CAM_D1_N	MIPI Data Lane 1 Negative
6	CAM_D1_P	MIPI Data Lane 1 Positive
7	GND	Ground
8	CAM_CK_N	MIPI Clock Lane Negative
9	CAM_CK_P	MIPI Clock Lane Positive
10	GND	Ground
11	CAM_IO0	Power Enable
12	CAM_IO1	LED Indicator
13	CAM_SCL	I2C SCL
14	CAM_SDA	I2C SDA
15	CAM_3V3	3.3V Power Input

3.2 Compatible Lenses

Option 1: D1326

Item	Specification
Horizontal FOV (HFOV)	97.43°
Vertical FOV (VFOV)	80.97°
Diagonal FOV (DFOV)	109.82°

Option 2: CW-8MP — See `camera_lens/` directory for full specifications.

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4. User Guide — Raspberry Pi 5

We provide pre-built driver and IPA packages. If you cannot find a package for your specific OS version or wish to build from source, contact us with your purchase order:

- zoujy@inno-maker.com
- support@inno-maker.com

4.1 Install Pre-built Driver and IPA

Step 1 — Download from GitHub:

```
https://github.com/INNO-MAKER/CAM-IMX577
```

```
cd CAM-IMX577
sudo chmod -R a+rwx *
```

Step 2 — Install the matching pre-built driver:

```
cd prebuild-driver-ipa
# Select the package matching your OS and kernel version
# Example for Bookworm Pi 5 kernel 6.12.75:
sudo tar -xzvf imx577_bookworm_pi5_k6.12.75+rpt-rpi-2712_20260616-134821.
cd imx577_bookworm_pi5_k6.12.75+rpt-rpi-2712_20260616-134821/scripts
sudo ./install.sh
sudo reboot
```

> **Supported packages:**
>
> | Package | OS | Kernel |
> |---|---|---|
> | `imx577_bookworm_pi5_k6.12.75+rpt-rpi-2712_20260616-134821.tar.gz` | Raspberry Pi OS Bookworm | 6.12.75+rpt-rpi-2712 |
> | `imx577_trixie_pi5_k6.12.75+rpt-rpi-2712_20260418-201631.tar.gz` | Raspberry Pi OS Trixie | 6.12.75+rpt-rpi-2712 |

Step 3 — Modify `config.txt` :

```
# /boot/firmware/config.txt (Pi 5)
camera_auto_detect=0
dtoverlay=imx577-overlay,cam0   # or cam1 if connected to CAM1 port
```

Step 4 — Test after reboot:

```
rpicam-hello --list-cameras
rpicam-hello -t 0
rpicam-still -o image.jpg --width 4056 --height 3040
rpicam-raw -t 5000 --width 4056 --height 3040 -o raw_image.dng
```

4.2 Offline libcamera Compilation (if 4.1 does not work)

```
sudo tar -zxvf libcamera_offline_build.tar.gz
cd libcamera_offline_build
sudo ./scripts/build_offline.sh
```

4.3 Driver Source Compilation (if 4.1 does not work)

```
sudo tar -zxvf driver_src_compiler.tar.gz
cd driver_src_compiler/scripts
sudo ./install_bookworm.sh   # or install_trixie.sh for Trixie
```

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5. User Guide — NVIDIA Jetson Orin Nano

5.1 Flash JetPack SDK 6.2

Follow the official NVIDIA getting-started guide:

```
https://developer.nvidia.com/embedded/learn/get-started-jetson-orin-nano-
```

5.2 Configure via Jetson-IO

Step 1 — Run the Jetson-IO Python script:

```
sudo /opt/nvidia/jetson-io/jetson-io.py
```

Step 2 — Select the **IMX477** option from the menu.

Step 3 — Reboot:

```
sudo reboot
```

5.3 Test the Camera

Verify detection:

```
v4l2-ctl --list-devices
```

Live preview (GStreamer):

```
gst-launch-1.0 nvarguscamerasrc ! \
  'video/x-raw(memory:NVMM),width=3840,height=2160' ! \
  nvvidconv ! xvimagesink
```

Full resolution capture:

```
gst-launch-1.0 nvarguscamerasrc num-buffers=1 ! \
  'video/x-raw(memory:NVMM),width=4032,height=3040' ! \
  nvjpegenc ! filesink location=capture.jpg
```

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6. Packing List

- 1 × IMX577 MIPI Camera Module
- 1 × 15-pin FPC cable
- 1 × 22-pin FPC cable

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7. Support

For technical support, source code requests, and product inquiries:

Channel	Contact
Technical Support	support@inno-maker.com
Sales	sales@inno-maker.com
Website	www.inno-maker.com
GitHub	github.com/INNO-MAKER/CAM-IMX577

> **Note**: Driver source code is available exclusively to customers who have purchased the CAM-IMX577. Please provide your purchase order when requesting source code access.